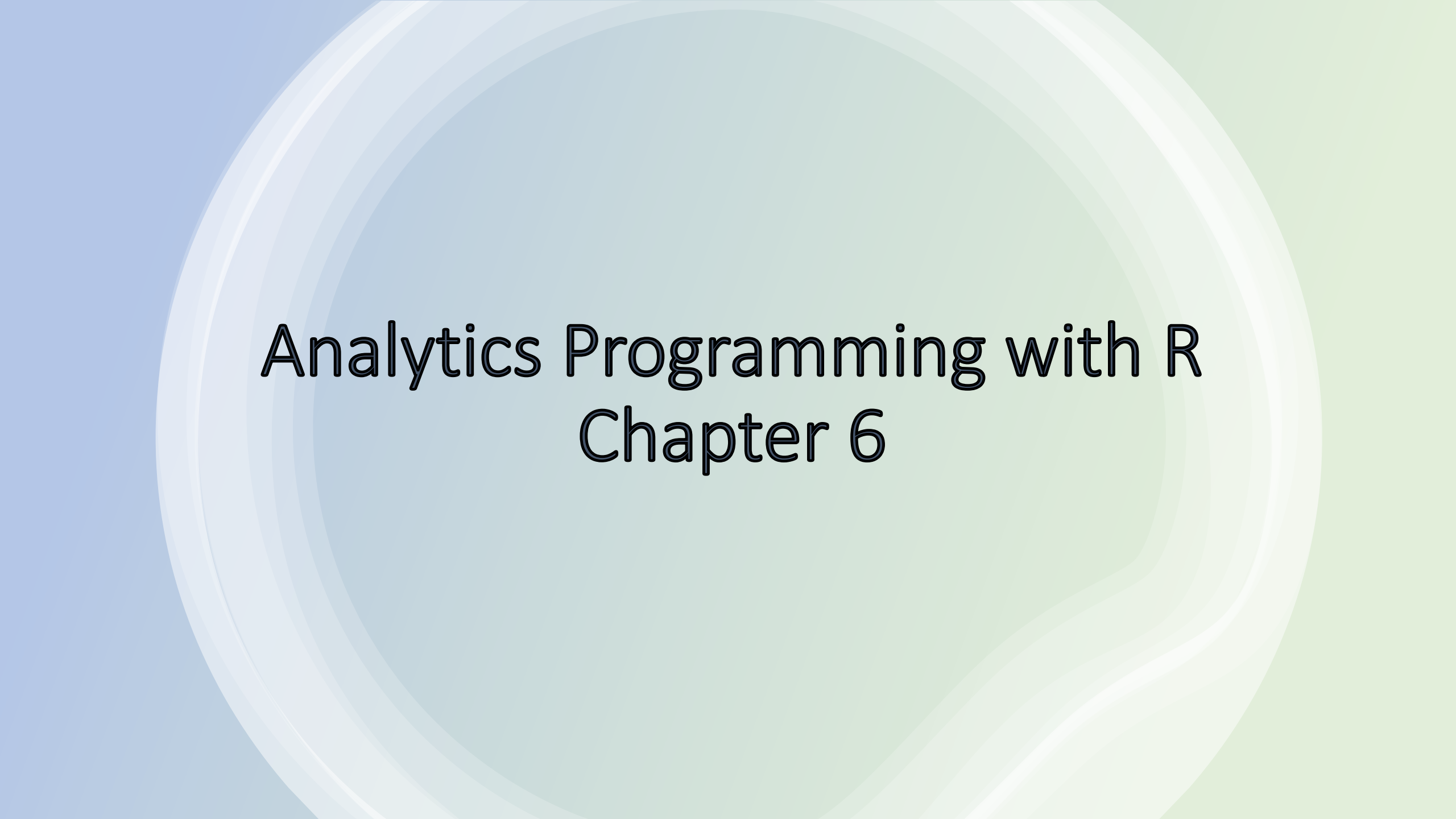
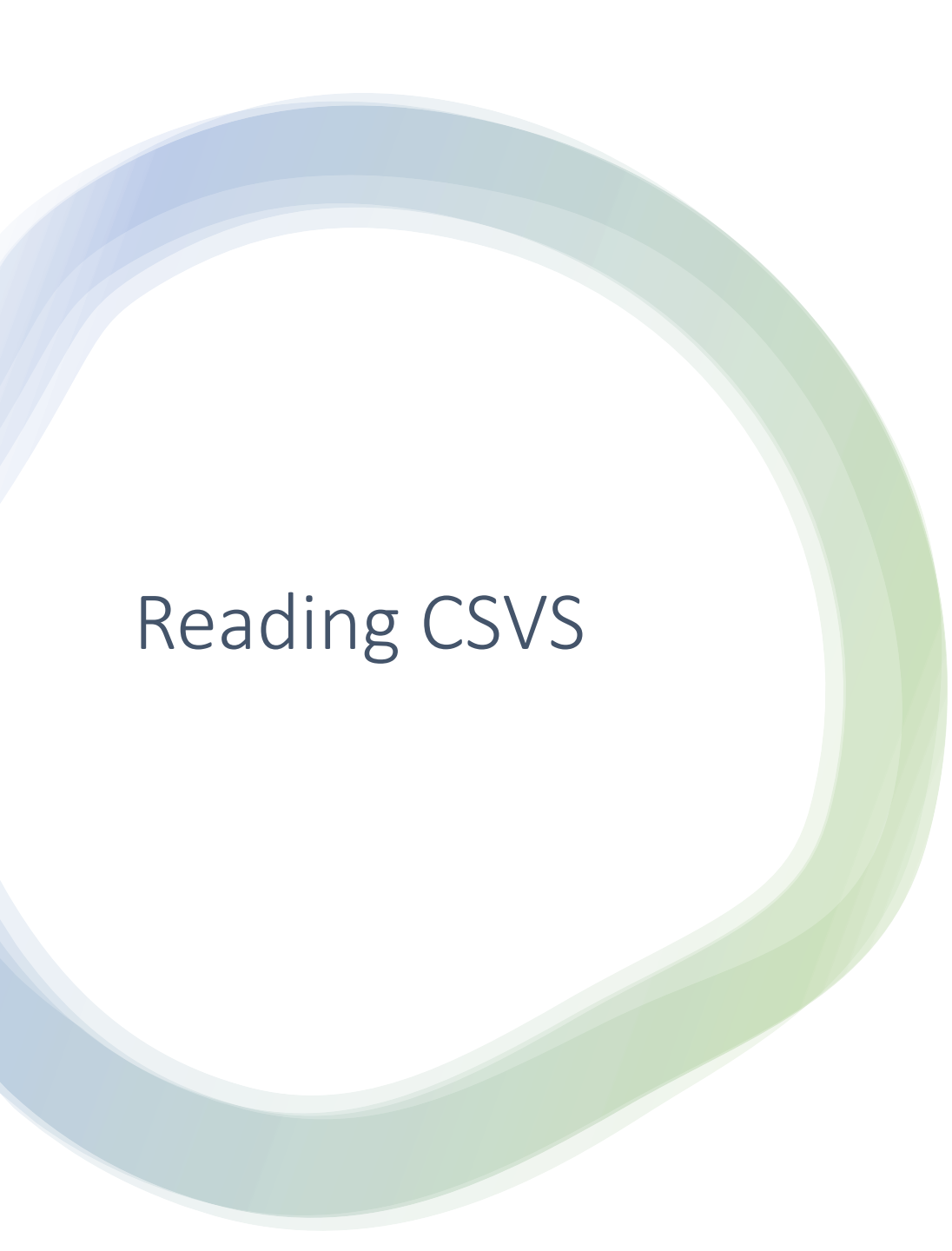




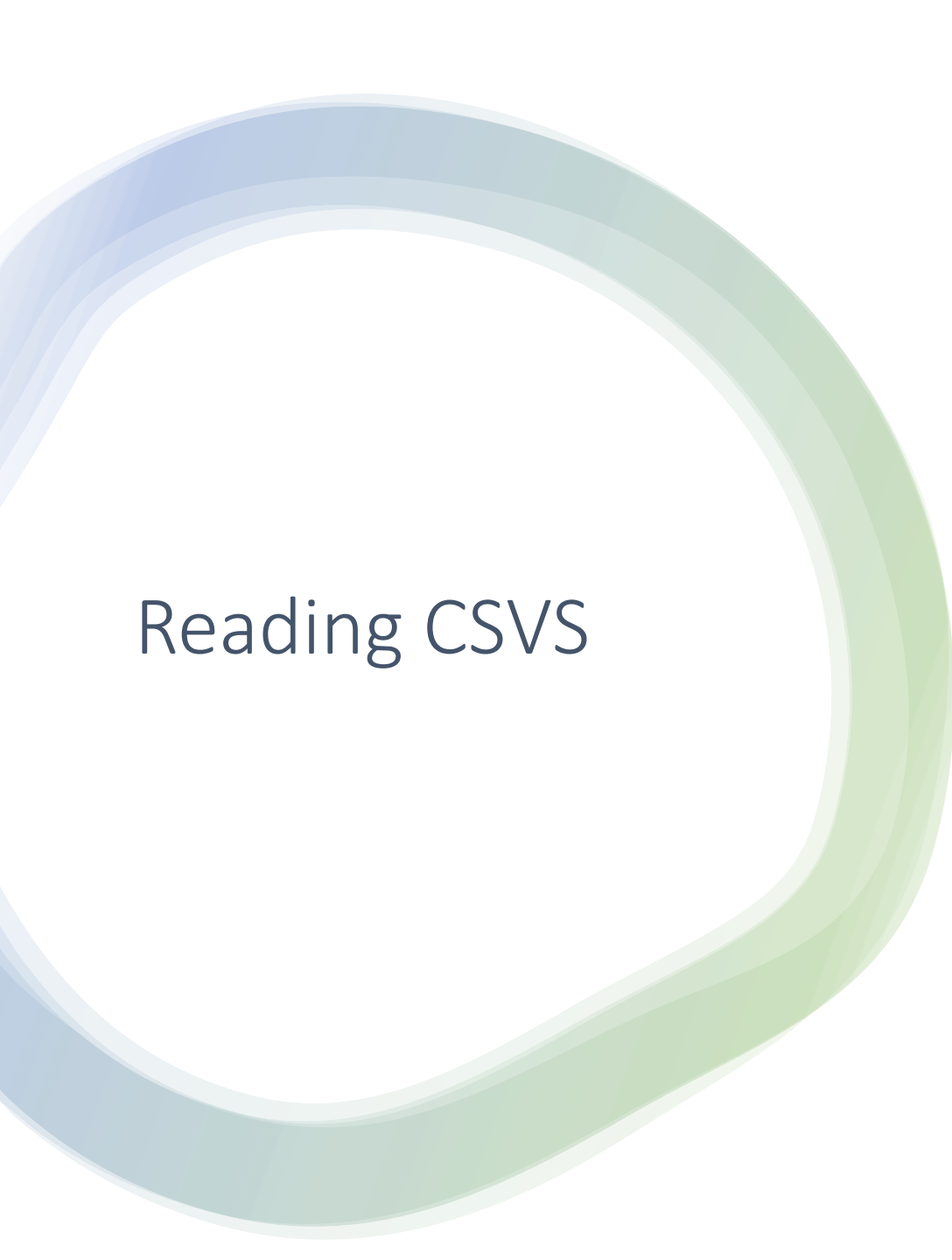
Analytics Programming with R

Chapter 6



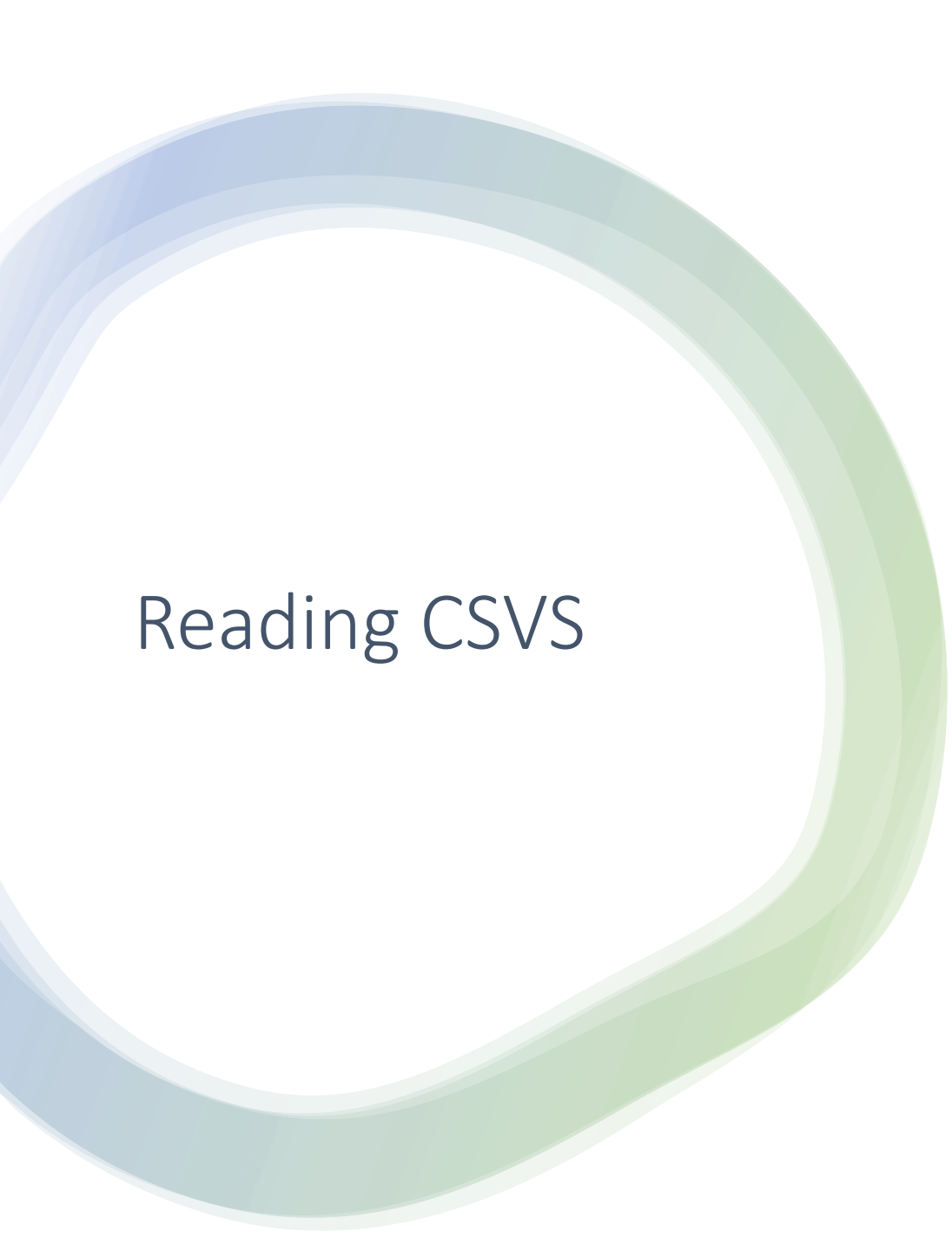
Reading CSVs

- In order to read data from a CSV use the `"read.table()"`
- The result of using `read.table` is a `data.frame`
- You can use the `head()` function to view the first few rows of data



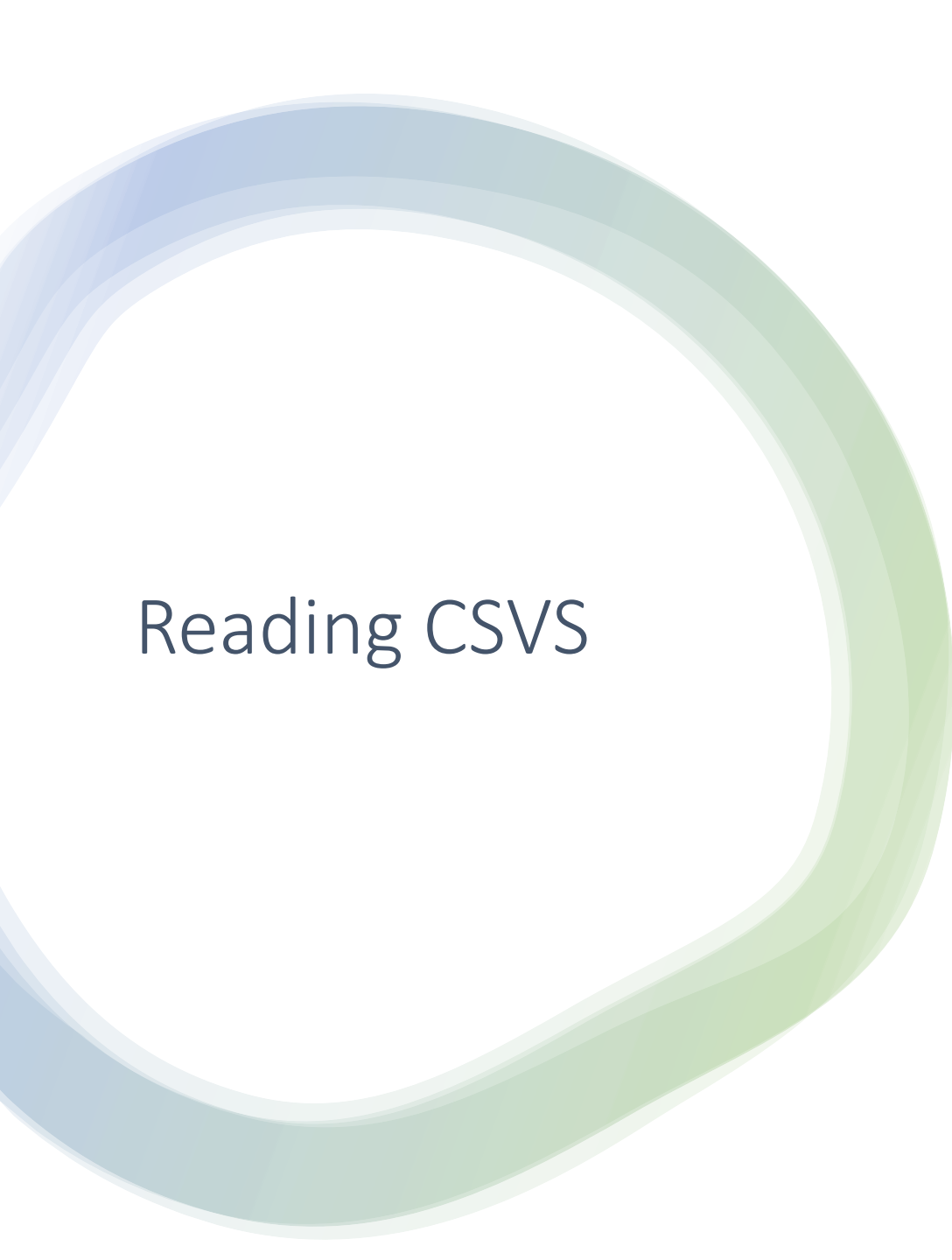
Reading CSVs

- function arguments can be specified without the name of the argument (positionally indicated), but specifying the arguments is good practice
 - Ex. `Read.table(file=theUrl, header = TRUE, sep=",")`
- The second argument, `header`, indicates that the first row of data holds the column names.
- The third argument, `sep`, gives the delimiter separating data cells. Changing this to other values such as `"\t"` (tab delimited) or `";"` (semicolon delimited) enables it to read other types of files.



Reading CSVs

- Another argument is `stringsAsFactors`.
- Setting this to `FALSE` (the default is `TRUE`) prevents character columns from being converted to factor columns.



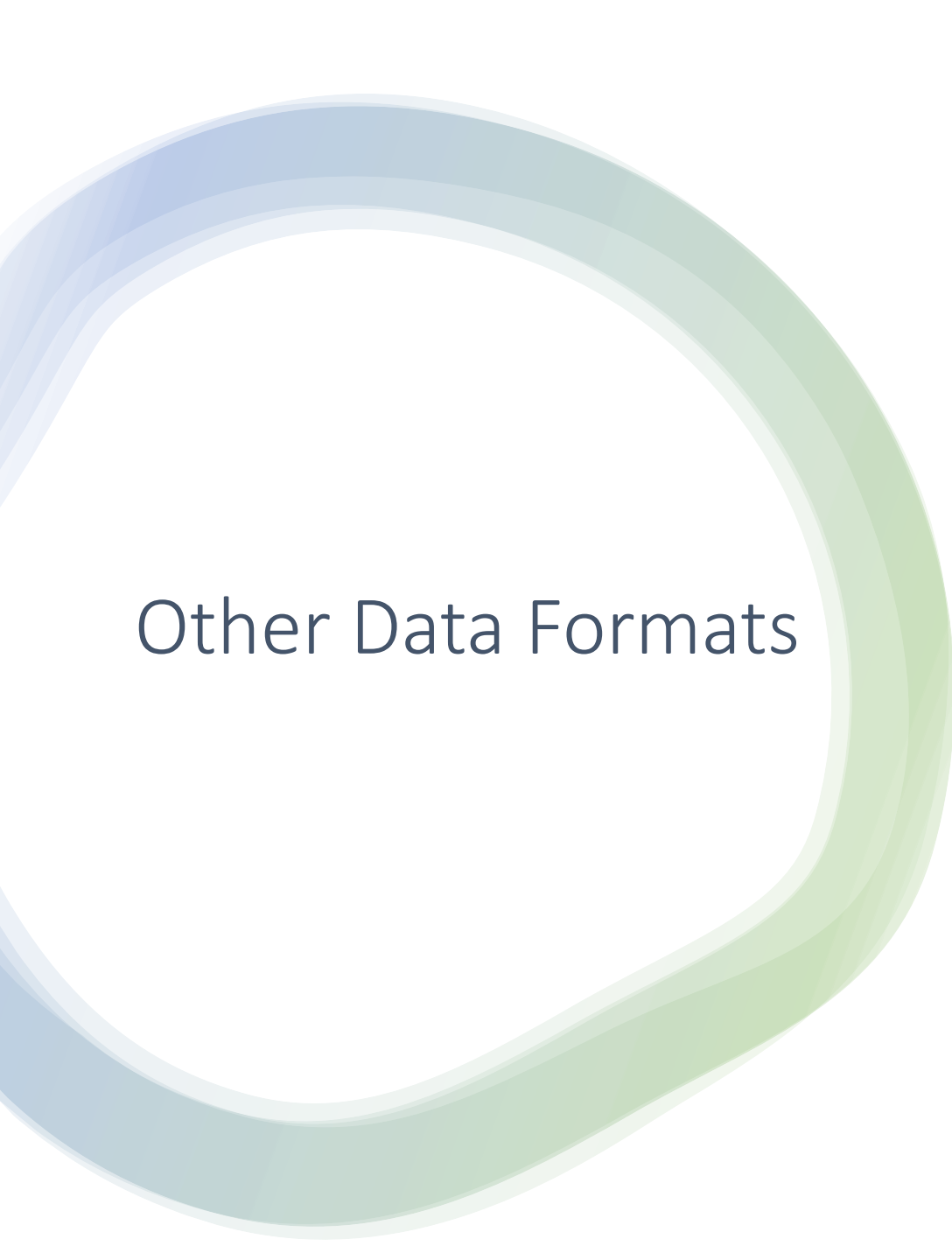
Reading CSVs

- The readr package has functions for reading text files
- Most common function is the read_delim function for CSV files
- Read_delim returns a tibble which is an extension of data.frame.
- **read_delim** is faster than **read.table**; it also removes the need to set stringsAsFactors to FALSE since that argument does not even exist.
- The functions **read_csv**, **read_csv2** and **read_tsv** are special cases for when the delimiters are commas (,), semicolons (;) and tabs (\t), respectively.



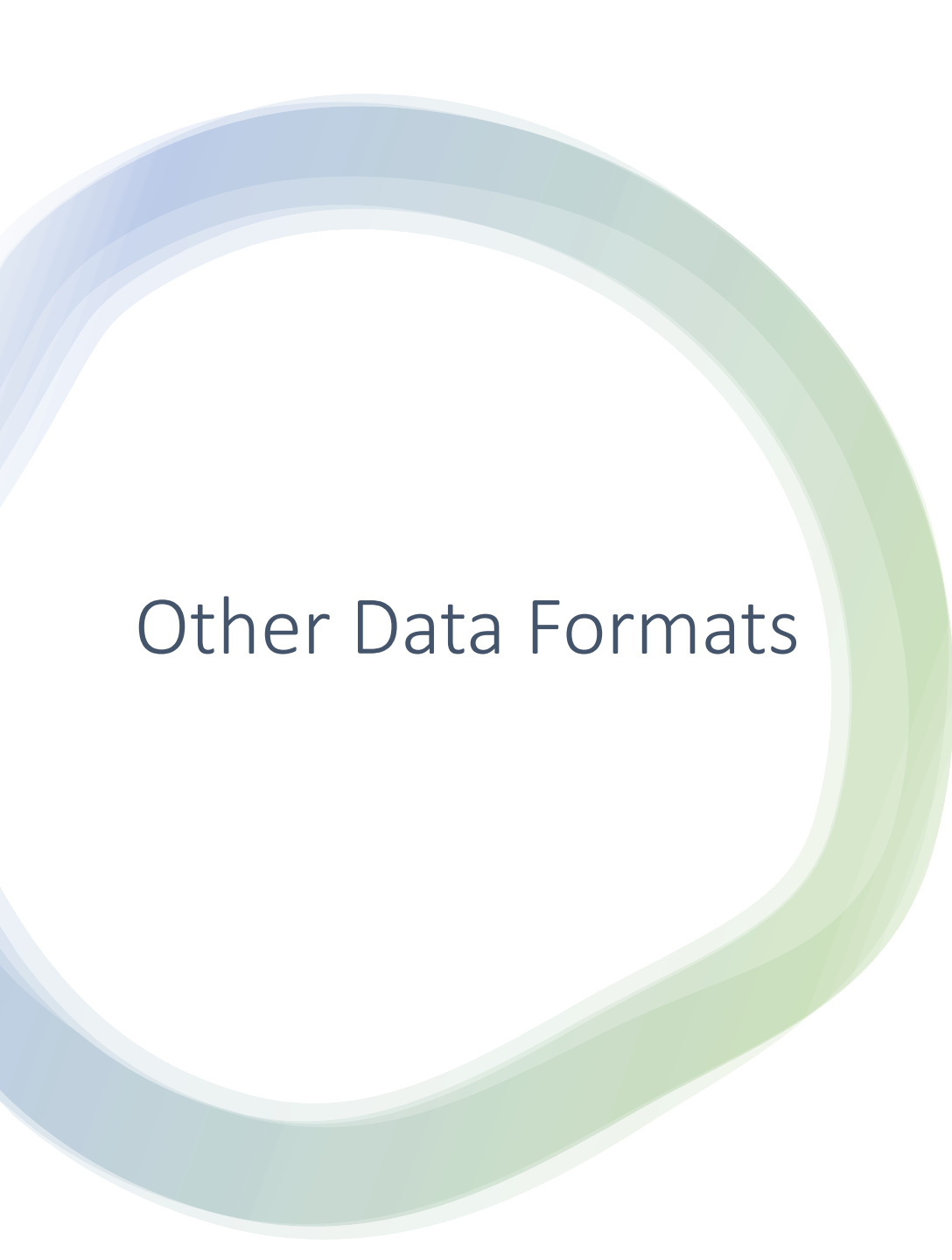
Excel Data

- The package **readxl**, by Hadley Wickham, makes reading Excel files, both `.xls` and `.xlsx`, easy.
- Main function is **read_excel**, which reads the data from a single Excel sheet.
- Unlike **read.table**, **read_delim** and **fread**, **read_excel** cannot read data directly from the Internet, and thus the files must be downloaded first



Other Data Formats

- Connecting to Databases is one of the most common ways to connect to data.
 - Typically, the connection is made using an ODBC connection to the database
 - There are many packages that will create the connections needed
- Getting data from other statistical tools such as SAS or SPSS
 - The “Foreign” package has tools that will let you connect to commonly used statistical tools



Other Data Formats

- Getting Data from websites
 - Reading HTML Tables
 - Scraping Data – rvest package is one of the packages used in R for data scraping.
- Reading Data from a JSON file
 - JSON (JavaScript Object Notation) file.
 - Used in programming APIs and webservices
 - Lightweight and easy to parse
 - Rjson and jsonlite used by R to read data from a JSON file